Harbin Institute of Technology (Shenzhen)
Project Report

Image Processing (COMP5033)

2025–2026 Spring Semester

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Project No.:	2	**Student Name:**	[YOUR NAME]
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1 Project Content

In this project we are given a 1024×1024 grayscale image (a yin-yang design with two cats) and we need to find its boundary using morphological image processing. The main steps are: first convert the grayscale image into a binary (black and white) image, then use morphological erosion to extract the boundary.

2 Method Description

2.1 Step 1: Binarization using Otsu's Method

To turn the grayscale image into binary I used **Otsu's thresholding**. The idea is to automatically find the best threshold value T that separates the image into foreground (bright) and background (dark) pixels.

Otsu's method tries every possible threshold from 0 to 255 and picks the one that maximises the variance between the two groups of pixels (foreground and background). A higher inter-class variance means the two groups are more separated.

```
Formula: σ²_between = w0 × w1 × (mean0 − mean1)² where w0 and w1 are the
proportions of pixels in each group.
```

For Figure_P2.jpg the algorithm found **T = 107**. Pixels brighter than 107 become white (1), the rest become black (0).

2.2 Step 2: Boundary Extraction using Erosion

Morphological **erosion** shrinks all white regions inward. For each pixel, I look at its 3×3 neighbourhood: if all 9 pixels are white, the pixel stays white; otherwise it becomes black.

Once I have the eroded image I can find the boundary by subtracting:

```
boundary = original_binary − eroded_binary
```

This leaves only the pixels that were removed by erosion – which are exactly the pixels on the edge of each white region.

I also do the same for the black regions (by inverting the binary image) to get the boundaries of the dark cat as well, then combine both results.

3 Experiment Results and Analysis

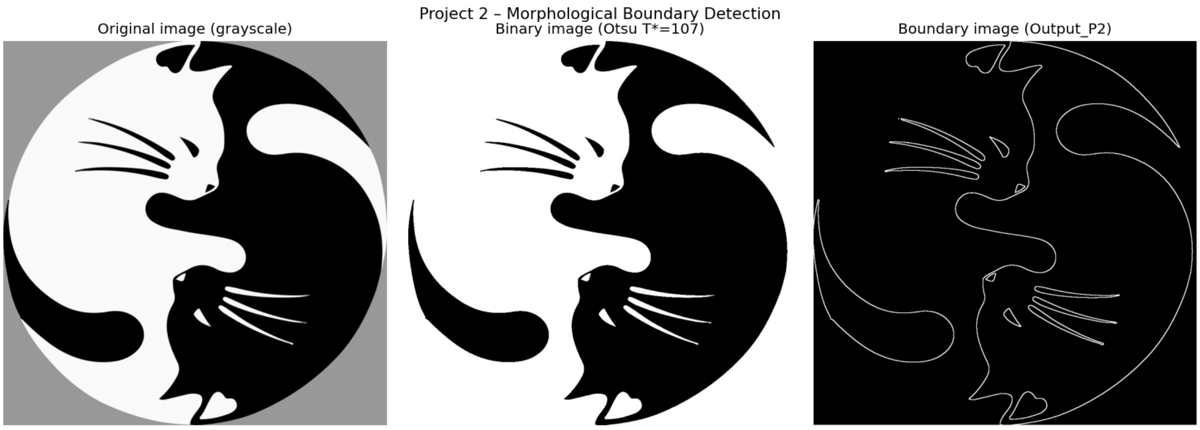

Figure 1. Left: original grayscale image. Centre: binary image (T=107). Right: boundary output (Output_P2.jpg).

Binarization result: Otsu's threshold of 107 works well for this image. The white cat body is correctly identified as foreground and the black cat as background. The gray border around the circle also becomes background.

Boundary result: The output shows a thin white line tracing all edges of the image – the outer circle, the S-curve between the two cats, the cat faces, whiskers, and tails. The total number of boundary pixels is 27,460.

Limitation: Very thin lines like the whiskers (thinner than 3 pixels) can disappear after erosion because the 3×3 neighbourhood removes them completely. A smaller structuring element would help in that case.

4 Summary

The hardest part was implementing Otsu's thresholding from scratch. I had to carefully compute the inter-class variance for each threshold value using the histogram. Once the binary image was correct, the erosion step was straightforward.

From this project I learned: (1) how to automatically binarize an image without manually choosing a threshold; (2) how erosion works and how subtracting an eroded image from the original gives the boundary; (3) that morphological operations are simple but very effective for shape analysis tasks.